

Co-Editing with AI: A Workflow That Emerged from Writing a Book

How a simple inline annotation technique solved an information management problem that was threatening to derail a 250-page manuscript

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How This Started

This essay was not planned. It emerged as a byproduct of writing a book — *Learning With AI: A Guide to Mastery, Curiosity, and the Art of Teaching Yourself* — at a point in the process where the manuscript had grown to roughly 250 pages across nineteen chapters, seven appendices, front matter, a conclusion, and an epilogue. The content was genuinely complex: multiple narrative threads, a technical spine running through several chapters, a set of appendices containing verbatim session transcripts, figures, and annotated bibliographies, and a growing set of companion essays and tools hosted on this website.

At that point, the standard editing question — “what needs to change?” — had become genuinely difficult to answer. Not because the answer was unclear, but because the volume and distribution of the problem made it hard to hold in mind at once. The diagnostic work and the corrective work kept getting tangled. Editorial suggestions made in conversation were easily lost. The gap between “what I think needs to happen” and “what I can actually act on in the file” was widening.

The workflow described here closed that gap. It did not require any new software, any automation, or any engineering beyond the tools already in use. It required one small innovation: moving the editorial commentary *inside the document itself*, in a form that could be compiled, read, and responded to in a structured way.

The Problem: Information Management at Scale

Writing a long book is, among other things, an information management problem. The content is not static. Chapters are revised, new material is added, references shift, voice inconsistencies accumulate, and the relationship between sections changes as the manuscript develops. Managing all of that is difficult enough when one person is doing it alone.

Adding an AI collaborator to the process creates a new version of the same problem. The AI has no persistent memory across sessions. Every session begins from scratch. The human has to re-establish context, re-state the goals, and re-explain the current state of the work before anything useful can happen. In a short document, this overhead is manageable. In a 250-page manuscript with a complex internal structure, re-establishing context at the start of every session is itself a significant cost.

The prior approach — describing problems in conversation, receiving suggestions as text, then manually finding the location in the document and deciding whether to act — worked adequately for small edits. For structural editorial work, where the question is not “fix this sentence” but “this entire section is pulling in the wrong direction and here is why,” it broke down. The suggestion and the content it referred to were in different places. Reading the suggestion required leaving the document. Returning to the document required remembering the suggestion. The cognitive overhead was real and growing.

The Innovation: Inline Diagnostic Annotation

The solution that emerged was conceptually simple. Before beginning any structural editorial pass on a chapter, a *diagnostic version* of that chapter is created — a copy of the original file, with “DIAG” prepended to the filename. Into this diagnostic copy, the AI inserts color-coded annotation blocks directly into the text, immediately adjacent to the content they describe.

Each annotation block has the same structure:

► **DIAG START — SHORT KEY**

Editorial note describing the issue, the recommendation, and the reason. Written in plain language. Self-contained — the reader does not need any other context to understand what is being said.

◄ **DIAG END — SHORT KEY**

The short key is a 2–4 word label that identifies the annotation uniquely: RESUME FEEL, KEEP STRONG, COMPRESS, CUT CANDIDATE, FIGURES OVERHAUL, and so on. The key appears in both the start and end tags so that the annotation is unmistakable in the compiled PDF.

The diagnostic version is compiled into a PDF and the human reads it as they would read any draft — sequentially, in context. The difference is that the editorial commentary is physically present in the document, immediately following or preceding the content it addresses. The human encounters the suggestion at exactly the moment they are reading the relevant passage. Nothing needs to be held in memory. Nothing needs to be cross-referenced.

The Review Cycle: A Pull Request for Prose

After reading the diagnostic PDF, the human responds with a structured reply that mirrors the format of a software pull request review. Each response references the annotation by its key and states a decision:

- **Accept** — make the change as suggested
- **Modify** — make a version of the change, with specific adjustments stated
- **Reject** — do not make this change, with a brief reason
- **Defer** — flag for a later pass, not this revision

The AI evaluates the responses before executing. If a decision seems to conflict with something established elsewhere in the manuscript, or if accepting one change creates a problem downstream, the AI flags it before proceeding. This is the “second opinion” step — not overriding the human’s decision, but checking it against the larger context before the edit is made.

Once the response is evaluated and any conflicts are resolved, the AI produces a clean version of the chapter — no annotation boxes, no diagnostic markers — with all accepted changes incorporated. This clean version replaces the diagnostic copy and becomes the new working file.

What the Workflow Actually Manages

To make the information management scale concrete: the manuscript this workflow was developed for contains the following elements, all of which are live and subject to revision:

- 19 chapter and front-matter files in LaTeX
- 7 appendix files, including two with verbatim AI session transcripts
- An annotated bibliography of 23 books across 5 thematic sections
- A table of contents with page number references that must be updated after any structural change
- 3 figures in the perceptron chapter (currently flagged for replacement)

- A set of companion essays and downloadable tools hosted at physicsexplorer.com, referenced by URL from the manuscript
- A running list of pending insertions, placement decisions, and phased content spanning multiple chapters

Without a structured workflow, tracking which of these elements is current, which is pending, and which has a known problem is itself a full-time job. The diagnostic annotation system does not solve all of this — it is specifically a tool for the editorial pass on individual chapters. But it makes that pass tractable in a way that prior approaches did not.

Why This Works: The Psychology of In-Context Feedback

There is a reason this approach works that goes beyond the mechanics. Editorial feedback that lives outside the document it refers to is easy to defer. The distance between the suggestion and the content creates friction — the same friction that makes it easy to close a comment thread in a Word document without acting on it, or to set aside a page of editorial notes and forget to return to them.

When the feedback is inside the document, that friction disappears. You cannot read past the annotation without encountering it. You have to engage with it or consciously decide to skip it. The annotation is not asking for your attention from a sidebar or a separate window. It is occupying the same space as the prose it is commenting on. That proximity changes how it feels to read and respond.

This is the same principle behind the Feynman technique: writing out what you know forces the gaps into view in a way that thinking about it does not. The inline annotation forces the editorial judgment into the reader's visual field at the exact moment it is most relevant.

What Is Not Required

It is worth being specific about what this workflow does not require, because the simplicity is part of what makes it useful.

It does not require any dedicated software. The annotation style used here is a custom LaTeX environment — a colored bordered box with specific typography — but the same effect could be achieved with any word processor's comment or highlight function. The key is that the annotation compiles or renders into the document, not into a sidebar.

It does not require automation or AI agents. The AI's role is entirely manual: read the content, form an editorial judgment, write the annotation, insert it at the right location. There is no pipeline, no automated processing, no system prompt engineering beyond the normal context of the conversation. The workflow is a prompting convention, not a technical system.

It does not require persistent AI memory. Because the annotation is in the document, and because the human responds by key name, the entire state of the editorial review is captured in the files themselves. The next session can begin by reading the previous diagnostic PDF and the human's response. Context is restored from the artifacts, not from memory.

The Broader Observation

This workflow emerged from a specific problem in a specific project. But the underlying principle generalizes: the most effective Human + AI collaboration tends to happen when the human's judgment and the AI's output are brought into the same space, rather than operating in parallel tracks that must be manually reconciled.

In the standard model of AI-assisted writing, the human asks for something, the AI produces it, and the human decides what to do with it. The two roles are sequential and separate. In the workflow described here, the AI's editorial judgment is embedded directly in the human's reading experience, and the human's response is structured in a way that the AI can evaluate before acting. The roles are interleaved, not sequential.

That interleaving is what makes the workflow feel like collaboration rather than delegation. The human is not handing off a task and receiving a result. The AI is not waiting for instructions and executing them. Both parties are engaging with the same content at the same time, each contributing what the other cannot: the AI contributing pattern recognition across the full manuscript, the human contributing the authority, judgment, and lived knowledge that the manuscript is ultimately about.

A Note on Emergence

This workflow was not designed. It was not planned at the start of the project or derived from a prior method. It emerged in the middle of an editorial session when the existing approach was clearly not working and a small innovation — “what if the annotation were inside the PDF?” — turned out to solve the problem directly.

That emergence is itself characteristic of the kind of Human + AI work this book is about. Neither party would have arrived at this particular solution alone. The human had the problem and the context. The AI had the technical capability to implement the annotation style and the ability to hold the full manuscript in view at once. The solution was the product of both, arrived at through the kind of iterative, conversational, genuinely collaborative session that the book argues is the highest use of the Human + AI pairing.

The irony — or perhaps the confirmation — is that the workflow will now be documented in the book itself, as one of the examples of what this kind of collaboration actually produces when it is working correctly.

Two Refinements Worth Adding

Since this workflow was first used, two small improvements have become apparent.

The summary map. After inserting all annotations and compiling the DIAG PDF, the AI should produce a brief index — a one-line description of each annotation keyed to its SHORT-KEY label. This gives the author a quick reference before reading the full chapter, so the overall editorial direction is visible at a glance before engaging with any individual annotation. In a long chapter with eight or ten annotations, knowing the shape of the feedback before reading it in detail reduces cognitive load considerably.

Keep your source files. The workflow depends on having access to the *LaTeX source* (the `.tex` file), not just the compiled PDF. If only a PDF is available, the AI must reconstruct the document structure from extracted text, which is lossier and slower. The practical rule is simple: always preserve your `.tex` files alongside the compiled PDFs, and hand the AI the source file when beginning a DIAG pass.

Appendix: The Standalone Prompt

What follows is a complete, self-contained prompt that can be used to execute a diagnostic editorial pass on any manuscript chapter. It requires no prior context — a new AI session can begin with this prompt and a chapter source file and produce a DIAG output in the style described in this essay. Copy it exactly. Modify only the filename and any project-specific framing.

What You Will Need Before You Start

- The **LaTeX source file** (`.tex`) for the chapter you want to diagnose. Upload it to the AI session.
- A clear sense of **what the chapter is supposed to do** — its role in the book and its central argument. You will state this briefly in the prompt.
- **Two pdflatex passes** available in the environment, or a local LaTeX installation if you are compiling yourself.

What You Will Get

- A **DIAG version** of the chapter (`.tex` and compiled `.pdf`) with color-coded annotation boxes inserted directly into the text
- A **summary map** listing every annotation by SHORT-KEY with a one-line description

- No changes to the actual chapter content — the DIAG version is read-only commentary

The Prompt

You are acting as a structural editor for a book manuscript. I am uploading a LaTeX chapter source file. Your job is to produce a DIAGNOSTIC version of that chapter using the workflow described below. Do not make any changes to the actual chapter content. The DIAG version is read-only editorial commentary only.

CONTEXT

Book title: [your book title]

Chapter: [chapter number and title]

What this chapter is supposed to do: [one or two sentences describing the chapter's role and central argument]

Known issues going in: [optional -- anything you already know needs work]

STEP 1 -- READ AND ASSESS

Read the chapter carefully as both a reader and a structural editor.

Identify sections or passages that fall into one of these categories:

KEEP -- strong content, serves the book's argument, no change needed

COMPRESS -- valid but takes more space than its contribution warrants

CUT -- does not serve the argument, reads like resume content, or belongs elsewhere

MOVE -- belongs in a different location in this chapter or the manuscript

EXPAND -- underdeveloped, needs more depth or narrative

FIGURES -- visual content that needs replacement or improvement

VOICE -- tone or register inconsistency

TRANSITION -- connection between sections is weak or missing

STEP 2 -- ADD THE PREAMBLE CODE

Add the following to the LaTeX preamble, before `\begin{document}`:

```
\usepackage{mdframed}
\usepackage{xcolor}
\newmdenv[headline=true, bottomline=true, rightline=true, leftline=true,
linewidth=2pt, linecolor=red!70, backgroundcolor=yellow!15,
innerleftmargin=12pt, innerrightmargin=12pt,
innertopmargin=8pt, innerbottommargin=8pt,
skipabove=10pt, skipbelow=10pt]{diagbox}
\newcommand{\diagnote}[2]{%
\begin{diagbox}%
{\small\bfseries\color{red!80} [»] DIAG START -- -- #1}\small #2}
{\small\bfseries\color{red!80} [«] DIAG END -- -- #1}%
}
```

```
\end{diagbox}%
}
```

STEP 3 -- INSERT ANNOTATIONS

Insert `\diagnote{SHORT-KEY}{note text}` blocks directly into the source, immediately before the content they describe. SHORT-KEY is a 2-4 word ALL-CAPS label unique to this annotation. The note text must be self-contained: name the problem, state the recommendation, give the reason in 2-4 sentences.

STEP 4 -- SAVE AND COMPILE

Save the annotated file as the original filename with DIAG prepended (e.g. BOOK2_16_Ch06_DIAG.tex). Compile with two pdflatex passes. Provide both the .tex and the .pdf.

STEP 5 -- PRODUCE THE SUMMARY MAP

After compiling, list every annotation in order: SHORT-KEY / category (KEEP, CUT, etc.) / one sentence describing what it flags. This is the author's quick-reference index before reading the full DIAG PDF.

AUTHOR RESPONSE FORMAT

After reading the DIAG PDF, the author responds by SHORT-KEY with one of:

ACCEPT -- make the change as suggested

MODIFY -- make a version of the change, with specific instructions

REJECT -- do not make this change (brief reason)

DEFER -- flag for a later pass

The AI evaluates responses before executing, flags any conflicts with other parts of the manuscript, then produces a clean version with all accepted changes incorporated.

That is the complete prompt. The only fields to fill in are the CONTEXT block at the top. Everything else is fixed and should be copied exactly. The workflow produces consistent, readable DIAG output regardless of chapter length, subject matter, or manuscript type.

Bruce DeSimas is the author of Learning With AI: A Guide to Mastery, Curiosity, and the Art of Teaching Yourself (in preparation) and the creator of physicsexplorer.com, an interactive AP Physics 1 learning platform. This essay was written during the editing process for that book, in June 2026.